

Geometric Postulates, Theorems, and Proofs

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Geometry is built from a few **undefined**
terms: point, line, and plane.

Point

A **point** represents an exact location and has no size.

- A

Point A has no length, width, or thickness — it is just a location.

Line

A **line** extends infinitely in two opposite directions.



The arrows on both ends show that the line never stops.

Plane

A **plane** is a flat surface extending infinitely in all directions.

Think of a plane like an endless flat sheet of paper with no thickness, extending forever in every direction.

Defined Terms

Defined terms are described using undefined terms.

Examples include:

- ▶ line segment
- ▶ ray
- ▶ angle
- ▶ collinear points

Postulate

A **postulate** (or axiom) is a statement accepted as true without proof.

Postulates are the starting building blocks of geometry — we do not prove them, we simply accept them.

Theorem

A **theorem** is a statement that must be proven using definitions, postulates, and previously proven theorems.

Unlike a postulate, a theorem needs logical justification before we can use it.

Proof

A **proof** is a logical argument that shows why a mathematical statement is true.

A proof connects given information to a conclusion using definitions, postulates, and theorems — step by step.

Common Postulates

Postulate 1

Through any two distinct points, there is exactly one line.

$A \bullet \quad \bullet B$

Only one straight line can pass through both A and B .

Postulate 2

If two lines intersect, they intersect at exactly one point.

Two distinct lines can cross at only a single shared point — never at more than one.

Postulate 3

Through any three non-collinear points, there is exactly one plane.

Non-collinear means the three points do not all lie on the same line. Together, they determine one flat plane.

Examples

Example 1

How many lines can pass through two distinct points A and B ?

Answer: Exactly one line, by the postulate:
“Through any two distinct points, there is exactly one line.”

Example 2

Lines m and n intersect. At how many points do they meet?

Answer: Exactly one point, by the postulate:
“If two lines intersect, they intersect at exactly one point.”

Example 3

Points P , Q , and R are non-collinear. How many planes contain all three?

Answer: Exactly one plane, by the postulate:
“Through any three non-collinear points, there is exactly one plane.”

Practice Exercises

Practice 1

Identify whether each is an **undefined term**, a **defined term**, a **postulate**, or a **theorem**.

label=(a) Point

lbbel=(b) Ray

lcbel=(c) Through any two distinct points there is exactly one line.

ldbel=(d) Vertical angles are congruent.

Practice 2

Classify each as **true** or **false**.

- =(a) A line has two endpoints.
- =(b) A plane extends infinitely.
- =(c) A point has length and width.
- =(d) A theorem must be proven.

Practice 3

Complete the statements.

label=(a) A line is determined by exactly _____
points.

label=(b) A ray has one _____.

label=(c) Three non-collinear points determine one
_____.

Practice 4

Name the defined term described.

- (a) Part of a line with two endpoints.
- (b) Part of a line with one endpoint.
- (c) Two rays sharing a common endpoint.

Practice 5

Determine whether the following statements are postulates or theorems.

label=(a) Through any two points there is exactly one line.

lbbel=(b) If two angles are supplementary to the same angle, then they are congruent.

lcbel=(c) Vertical angles are congruent.

Practice 6

State the reason for each statement.

- =(a) Two points determine exactly one line.
- =(b) Vertical angles are congruent.
- =(c) Three non-collinear points determine one plane.

Practice 7

Identify the undefined terms used to define each.

=(a) Line segment

=(b) Ray

=(c) Angle

Practice 8

Arrange the following proof steps in the correct order.

label=(a) Apply the theorem.

lbbel=(b) State the given.

lcbel=(c) Write the conclusion.

ldbel=(d) Use the appropriate postulate.

Activities

Activity 1

Identify whether each is an **undefined term**, a **defined term**, a **postulate**, or a **theorem**.

label=(a) Plane

lbbel=(b) Line segment

lcbel=(c) If two lines intersect, they intersect at exactly one point.

ldbel=(d) The sum of the angles in a triangle is 180° .

Activity 2

Classify each as **true** or **false**.

- =(a) A ray has two endpoints.
- =(b) A line extends infinitely.
- =(c) A plane has thickness.
- =(d) Postulates are accepted without proof.

Activity 3

Complete the statements.

=(a) A ray has one _____.

=(b) A line extends infinitely in _____
directions.

=(c) Two distinct points determine one _____.

Activity 4

Name the defined term described.

- =(a) A flat surface extending infinitely.
- =(b) An exact location.
- =(c) A straight path extending infinitely in opposite directions.

Activity 5

Determine whether the following statements are postulates or theorems.

label=(a) Through three non-collinear points there is exactly one plane.

lbbel=(b) The base angles of an isosceles triangle are congruent.

lcbel=(c) If two lines intersect, they intersect at one point.

Activity 6

State the reason for each statement.

- label=(a) Three non-collinear points determine one plane.
- lbbel=(b) Base angles of an isosceles triangle are congruent.
- lcbel=(c) Two intersecting lines meet at exactly one point.

Activity 7

Identify the undefined terms used to define each.

=(a) Plane

=(b) Line

=(c) Angle

Thank you!